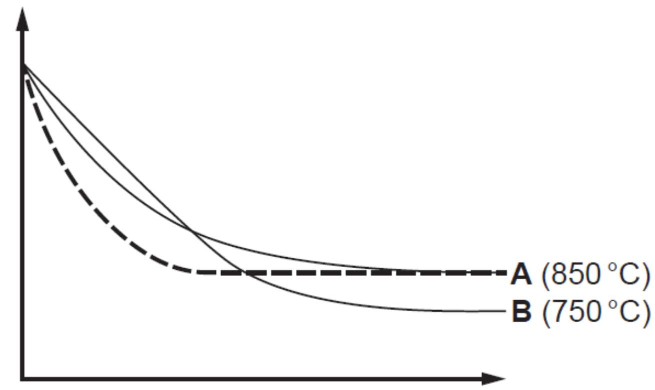
Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
9.	(a)			award (1) for any of following • H_2SO_4 loses protons / H^+ ions and NH_3 gains protons / H^+ ions • H_2SO_4 is a proton donor and NH_3 is a proton acceptor • NH_3 accepts a proton from H_2SO_4 neutral answer – reference to salt formation			1	1		
	(b)			moles $(NH_4)_2SO_4 = \frac{1.86}{132.18} = 0.0141$ (1) moles NaOH = 0.0282 (1) concentration $= \frac{0.0282}{0.0267} = 1.06$ mol dm^{-3} (1)		3		3	1 1	
	(c)	(i)		if a system at equilibrium is subjected to a change then the position of equilibrium will shift to minimise/oppose that change	1			1		
		(ii)		student **correct** since more moles of ammonia present at equilibrium at higher temperature (1) system opposes increase in temperature by shifting equilibrium position to the left / in the endothermic direction (1) neutral answer – reference to the 'endothermic <u>side</u>' if **incorrect** award no marks accept converse answer student **correct** since fewer moles of ammonia present at equilibrium at lower temperature (1) system opposes decrease in temperature by shifting equilibrium position to the right / in the exothermic direction (1)	1		1	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		 (1) catalyst does not affect the position of equilibrium / number of moles present at equilibrium but equilibrium reached more quickly / reaction is faster (1)	1	1		2		
	(d)			1.26 / 1.3		1		1	1	
	(e)			nitrogen more electronegative than hydrogen / nitrogen and hydrogen have different electronegativities (1) this results in polarity / unequal electron distribution in the bond (1)	1		1	2		
				Question 9 total	**4**	**5**	**3**	**12**	**3**	**0**